COLLEGE MINI PROJECT PRESENTATION

Break-Even Point Predictor & Feasibility Engine

PROJECT METADATA

Domain: FinTech / Financial Engineering & Web Systems

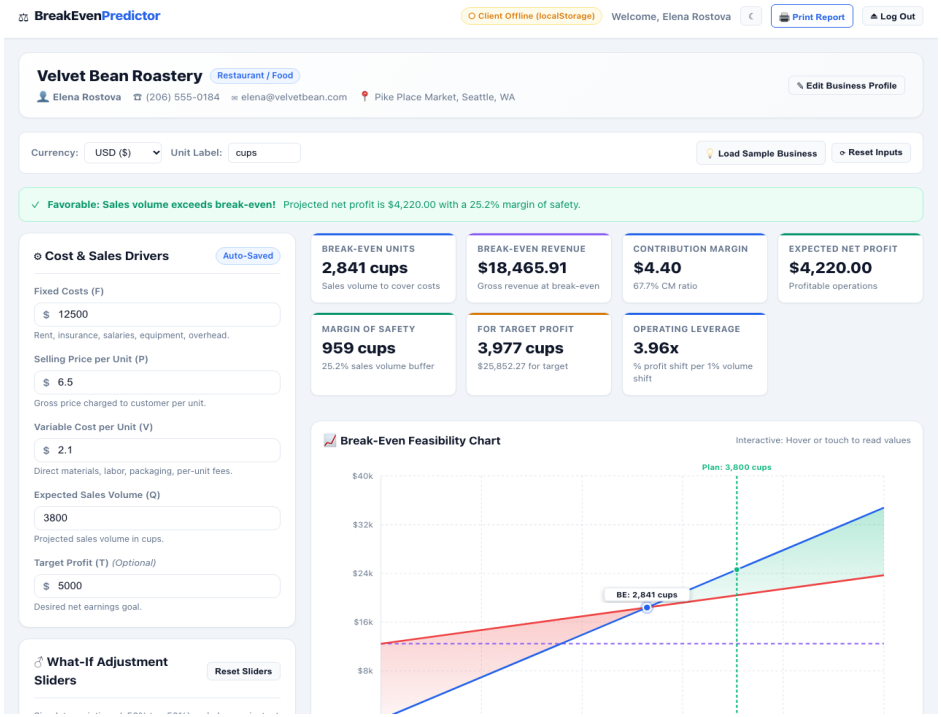
Developer: arjun2007arush2011-max

Architecture: Zero-Dependency Vanilla ES5 + Multi-Runtime REST

Database: SQLite3 Relational DB + LocalStorage Fallback

Live URL: https://arjun2007arush2011-max.github.io/break_even_point/

GitHub: https://github.com/arjun2007arush2011-max/break_even_point



BACKGROUND & MOTIVATION

Problem Statement & Project Objectives

THE PROBLEM STATEMENT

• **High Failure Rate:** Over 65% of startup failures stem from premature cost miscalculations and hidden overhead.

• **Static Spreadsheet Errors:** Traditional Excel files lack interactive dynamic curves and mobile responsiveness.

• **Blind Pricing Decisions:** Entrepreneurs often underprice without analyzing variable cost compression on margins.

• **No Sensitivity Simulation:** Business owners struggle to forecast the impact of 15% cost inflation or rent spikes.

KEY PROJECT OBJECTIVES

✔ **Zero-Dependency Web Platform:** Completely self-contained vanilla ES5 implementation running without CDNs or frameworks.

✔ **Dynamic SVG Graphics:** Visual curves for Total Revenue, Total Cost, Fixed Costs, and profit/loss zones.

✔ **Real-Time What-If Engine:** Sliders (-50% to +50%) and sensitivity tables to test cost and price variations.

✔ **Multi-Runtime REST Integration:** Idiomatic backends in Python, Ruby, and Node.js with SQLite3 persistence.

✔ **Executive Feasibility Reports:** Formatted printable reports for stakeholders with company letterhead.

SYSTEM ARCHITECTURE

Full-Stack System Architecture: Frontend & Backend

FRONTEND CLIENT TIER (Browser)

- **Vanilla ES5 Standard:** 100% universal compatibility across all browser engines without build steps or Babel.
- **Responsive Two-Column Grid:** Adapts smoothly from desktop to mobile screens.
- **Dark / Light Theming:** Driven by CSS Custom Properties with localStorage persistence.
- **Client-Side Crypto:** Passwords hashed via crypto.subtle (SHA-256) prior to storage.
- **Modular Namespaces:** AuthModule, ProfileModule, MetricsModule, ChartModule, ApiClient.

BACKEND SERVER & DATABASE TIER

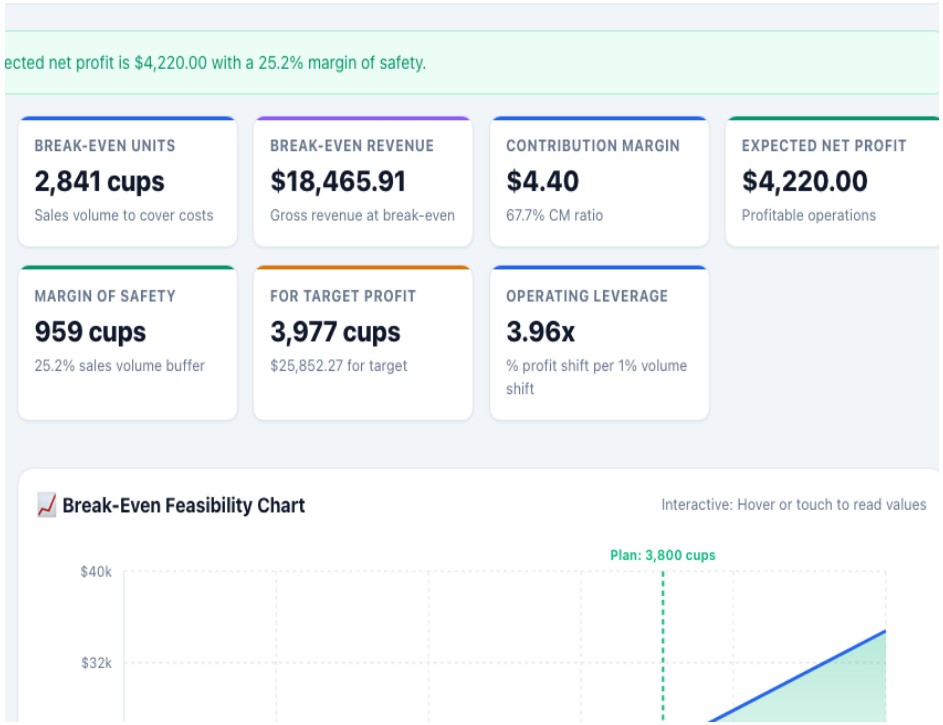
- **Triple-Runtime Servers:** Python 3 (http.server), Ruby (WEBrick), and Node.js. Zero pip/npm dependencies.
- **Universal Launcher (start.sh):** Auto-detects installed runtime and binds port 8000.
- **SQLite3 Database (breakeven.db):** Tables for Users, Profiles, and Cost Inputs.
- **REST API Contracts:** Endpoints for /api/auth/*, /api/profile, /api/inputs, /api/calculate.
- **Server Mathematical Verification:** Confirms calculation parity with client engine.

COMMUNICATION PROTOCOL & RESILIENT DUAL-SYNC BRIDGE (Frontend ■ Backend)

- **Debounced REST API Sync:** Client debounces user inputs (300ms) and sends JSON payloads with Bearer token authentication to /api/inputs and /api/profile.
- **Dual Persistence Resilience:** ApiClient pings /api/health. When server is active, records persist to SQLite3; if offline or on GitHub Pages (file://), it gracefully falls back to localStorage with zero downtime.

VISUALIZATION ENGINE

Interactive SVG Chart & Visual Capabilities



CUSTOM COORDINATE MAPPING ENGINE

- **Zero External Chart Libraries:** Custom SVG mathematical projection avoiding heavy Chart.js or D3.js libraries.
- **Loss & Profit Polygons:** Fills red loss zone (0 to BE) and green profit zone (BE to Max Volume) dynamically.
- **Break-Even Point Marker:** Displays intersection point with a pulsing halo and coordinate callout badge.
- **Expected Plan Line:** Vertical dashed indicator showing planned sales volume versus break-even.
- **Touch & Hover Crosshairs:** Real-time cursor reading volume, revenue, costs, and profit at 60fps.

FINANCIAL MODEL & ALGORITHMS

Mathematical Formulation & Financial Algorithms

CORE ACCOUNTING FORMULAS

- **Contribution Margin (CM):** CM = Selling Price (P) - Variable Cost (V)
- **Contribution Margin Ratio:** CM% = (P - V) / P
- **Break-Even Volume:** Q_BE = Fixed Costs (F) / (P - V)
- **Break-Even Revenue:** R_BE = Q_BE × P = F / CM%
- **Net Operating Profit:** Profit = Q × (P - V) - F
- **Margin of Safety (MOS):** MOS = (Q - Q_BE) / Q × 100%
- **Operating Leverage (DOL):** DOL = (Q × CM) / Profit
- **Target Profit Units:** Q_Target = (F + Target) / CM

ALGORITHMIC GUARDS & INTEGRITY

✔ **Structural Loss Detection (P ≤ V):** Instantly alerts if price is less than or equal to variable cost. Prevents division-by-zero exceptions.

✔ **Sub-Break-Even Operating Alert:** Dynamically signals financial danger when planned sales volume Q < Q_BE.

✔ **Floating-Point Normalization:** Sanitizes user numeric inputs and eliminates IEEE 754 floating-point inaccuracies.

✔ **Server Verification Parity:** Backend /api/calculate endpoint independently validates mathematical formulas.

SIMULATION & SENSITIVITY

What-If Sensitivity Simulations & Scenario Planning

REAL-TIME WHAT-IF SLIDERS

• **Bidirectional Sliders (-50% to +50%):** Instant adjustment of Price, Variable Cost, and Fixed Costs.

• **Live Variance Badges:** Computes real-time percentage shifts and monetary changes against base parameters.

• **Real-Time SVG Morphing:** Updates coordinate curves at 60fps, shifting the break-even intersection point dynamically.

• **Single-Click Baseline Reset:** Restores saved base parameters without page reloads.

STRESS TESTING & REPORT GENERATION

• **5-Tier Sensitivity Matrix:** Evaluates BE Volume shifts across -20%, -10%, 0%, +10%, and +20% cost driver changes.

• **Volume Scenario Analysis:** Models Pessimistic (-25%), Base, and Optimistic (+25%) market outcomes.

• **Executive Printable Report:** Dedicated @media print stylesheet for clean, publication-ready PDF exports.

• **Decision Support Warnings:** Highlights when fixed costs exceed safe coverage thresholds.

CONCLUSION & IMPACT

Project Results, Live Deployment & Academic Scope

KEY DELIVERABLES & OUTCOMES

✔ **Live Production Deployment:** Hosted globally on GitHub Pages at https://arjun2007arush2011-max.github.io/break_even_point/

✔ **Zero Dependencies:** Operates without npm packages, external build tools, or CDN dependencies.

✔ **High Performance:** Sub-millisecond mathematical calculations and smooth 60fps SVG chart recalculations.

✔ **Print Feasibility Report:** Clean @media print executive feasibility report for stakeholders.

✔ **SEO Optimization:** Integrated robots.txt, sitemap.xml, and OpenGraph social metadata.

ACADEMIC IMPACT & FUTURE SCOPE

• **Cross-Disciplinary Integration:** Merges software engineering, managerial cost accounting, database design, and interactive computer graphics.

• **Multi-Product Sales Mix:** Future expansion to calculate weighted average contribution margins for multi-item inventory catalogs.

• **Monte Carlo Risk Modeling:** Simulating demand curve fluctuations and price elasticity for probabilistic risk assessment.

• **GitHub Repository:** https://github.com/arjun2007arush2011-max/break_even_point